

Protective effect of *Premna tomentosa* extract (L. verbanacae) on acetaminophen-induced mitochondrial dysfunction in rats

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Abstract

Allurement of herbs as health beneficial foods (physiologically functional foods) and as a source material for the development of new drugs, has led to greater furtherance in the study of herbal medicines during recent years. Plant extracts are being utilized to treat a wide variety of diseases like hepatotoxicity. *Premna tomentosa* is one such medicinal plant used widely in Indian ayurvedic medicine for the treatment of liver disorders. This study appraised the effectiveness of *P. tomentosa* leaf extract in protecting the liver against mitochondrial damage induced by acetaminophen, since mitochondrial injury has been investigated as a potential initiator of hepatotoxicity. Normal Wistar strain rats were pre-treated with *P. tomentosa* extract (750 mg/kg, orally) for 15 days and then intoxicated with acetaminophen (640 mg/kg, orally). Mitochondria were isolated from liver of experimental animals and assessed for the levels of lipid peroxide products, GSH and mitochondrial enzymes (isocitrate dehydrogenase, α -keto glutarate dehydrogenase, succinate dehydrogenase, malate dehydrogenase, NADH dehydrogenase and cytochrome-C-oxidase). The levels of Lipid peroxidation products were increased and the levels of the other assessed parameters were significantly decreased in hepatotoxicity induced animals. Whereas, the levels were brought back to normal in *P. tomentosa* pre-treated rats, which shows the protective effect of the extract against mitochondrial damage. Presence of anti-oxidant compound D-limonene (58%) in *P. tomentosa* leaves, which is known to enhance conjugation of toxic metabolites by maintaining liver GSH concentrations may explain the hepatoprotective property of the extract. (Mol Cell Biochem 272: 171–177, 2005)

Key words: *Premna tomentosa*, mitochondria, acetaminophen, liver, limonene

Introduction

Pathogenesis of drug or toxin induced liver injury usually involves participation of toxic metabolites that directly affect the biochemistry of the cell, which is due to the occurrence of intracellular stress in the form of biochemical events such as oxidative stress, redox changes, covalent binding, lipid peroxidation, etc. The consequent functional effects are, either direct effect on mitochondria or effects on other organelles which send signals to mitochondria to recruit

participation in the death process [1]. Mitochondrial injury has been proposed to play a key role in the initiating events that leads to cytotoxic effects of several xenobiotics like acetaminophen.

Acetaminophen was shown to impair mitochondrial respiration in mice as early as 1-h post-dosing, while gross cellular necrosis were not observed until 3 h [2], which strongly suggest a role for mitochondrial dysfunction in the initiation of acetaminophen-induced hepatotoxicity. As well, acetaminophen-induced liver injury is a commonly used

model for screening of antihepatotoxic or hepatoprotective activity of drugs [3]. Since mitochondrial integrity is crucial for cell viability, the protective effect of *Premna tomentosa* extract was assessed in acetaminophen-induced mitochondrial dysfunction in rat liver.

P. tomentosa L. (Verbanaceae) is a medicinal plant used extensively for the treatment of various disorders. It is a moderate-sized deciduous tree with shoots, leaves and inflorescence densely clothed with a tawny yellow stellate tomentum. Bark is light grayish-brown, like that of teak and flowers are greenish yellow. Indian siddha medicine claims a lasting cure for hepatic disorders through oral administration of an extract of *P. tomentosa* [4]. In traditional medicine, *P. tomentosa* has also been used to treat stomach disorders and diarrhoea. The leaves are said to possess diuretic properties [5]. Earlier studies have shown that the methanolic extract of *P. tomentosa* leaves afforded significant protection against acetaminophen-induced hepatotoxicity in rats [6] by its antioxidant property [7]. The leaves also possess lipid-lowering effect [8] and immunomodulatory activity in lymphocytes [9] and macrophages [10].

Preliminary phytochemical screening of the leaves revealed the presence of the following compounds—D- and DL-limonene (57.8%), beta-caryophyllene (17.2%), cadalene type sesquiterpene (7.8%), sesquiterpene tertiary alcohol (5.6%) and aditerpene (5.5%). [11]. The major compound present in the leaves (58%) are limonene which is a monocyclic terpene consumed by humans predominantly as an ingredient of traditional food, e.g. citrus fruit, carrots, coffee, orange and nutmeg [12]. Investigations on the evaluation of the biological activities of D-limonene have revealed the antioxidant property of the compound [13]. D-Limonene is reported to induce glutathione (phase-II detoxification) in mice liver challenged with paracetamol/acetaminophen [14] which supports our previous report on modulating property of *P. tomentosa* extract on oxidative stress in acetaminophen intoxicated rats [7]. Since pathogenesis of liver injury promote a variety of chemical reactions including oxidative stress with direct effect on organelles like mitochondria, the present study was designed to examine the effect of *P. tomentosa* extract (which possess antioxidant compound limonene) on mitochondrial function in acetaminophen-induced hepatotoxicity in rats.

Materials and methods

Chemicals

Acetaminophen was purchased from Sigma, St. Louis, MO. Other chemicals used for the investigation were of analytical grade.

Animals

Adult male albino rats of Wistar strain were obtained from Fredrick Institute for Plant Protection and Toxicology (FIP-PAT), Padappai, Chennai, India. They were acclimatised to laboratory condition for 10 days and were fed a commercial pelleted rat chow (Hindustan Lever Limited, Bangalore, India) and water *ad libitum*.

Preparation of plant extract

P. tomentosa leaves were collected from a rural area near Palayamkottai, India and authenticated by Dr. Usman Ali, Drug Research Scheme, Multidisciplinary Unit, Central Research Institute for Siddha, Chennai 106, India. The powdered leaves were extracted with methanol to a syrupy mass. The last traces of the solvent were removed in vacuum with an approximate yield of 12.4%.

Study design

The animals were divided into four groups comprising of six animals each. Group 1 received olive oil and the vehicle 50% propylene glycol, orally. Group 2 was given olive oil for 15 days and acetaminophen (640 mg/kg, orally) administered 24-h post-treatment of the last dose. Group 3 was treated similar to Group 2 except that the plant extract (750 mg/kg, orally) [6] was administered instead of olive oil. Group 4 was given plant extract alone for 15 days. Animals were sacrificed by cervical decapitation, 24 h after the last dosage of drug or vehicle administration. The liver (homogenized in ice-cold isotonic sucrose) obtained from the animals were used for the estimation of the assessed parameters.

Mitochondrial preparation

Differential centrifugation was employed to separate mitochondria in liver, according to the method of Johnson and Lardy [15].

Estimation of Lipid peroxidation products and GSH

Mitochondrial lipid peroxidation products and reduced glutathione were estimated by the method of Okhawa *et al.* [16] and Moron *et al.* [17] respectively.

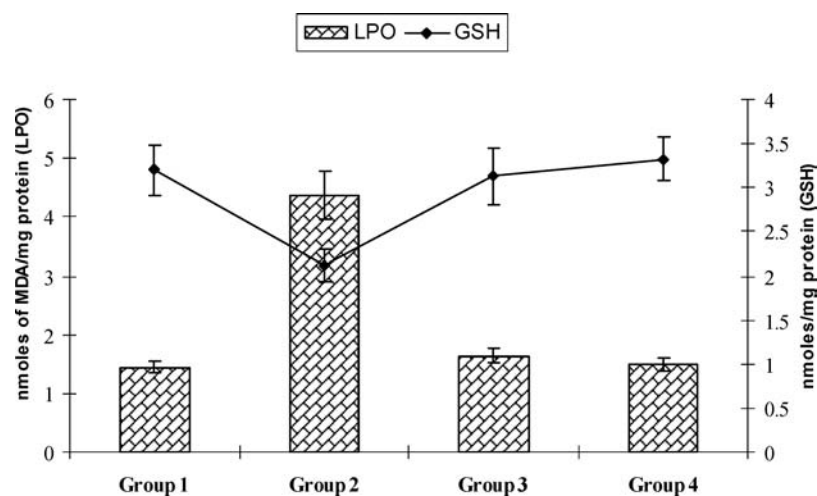


Fig. 1. Statistical comparisons: $p < 0.001$ for Group 2 vs. Group 1 and Group 3 vs. Group 2.

Assay of mitochondrial enzymes

The amount of α -keto glutarate formed were determined by the method of Bell and Baron [18] for estimation of isocitrate dehydrogenase (IDH) levels. The levels of α -keto glutarate dehydrogenase (α -KGDH) was assayed according to the colorimetric determination of ferrocyanide produced by the decarboxylation of α -ketoglutarate with ferricyanide as electron acceptor [19]. Succinate dehydrogenase (SDH) and cytochrome-C-oxidase were estimated according to the methods of Slater and Bonner [20] and Pearl *et al.* [21] respectively. Malate dehydrogenase (MDH) and NADH dehydrogenase (NADH DH) were quantified from the amount of NADH oxidised, by the methods of Mehler *et al.* [22] and Minakami *et al.* [23] respectively.

Analysis of data

The results were expressed as mean $\pm$ S.D. and all statistical comparisons were made by Student's *t*-test and is expressed as *p* (probability) values.

Results

Effect of *P. tomentosa* extract on levels of lipid peroxidation products and GSH

Treatment of animals with acetaminophen resulted in a significant increase in lipid peroxidation (3.01-fold), as measured by the levels of lipid peroxidation products. The levels of the reducing equivalent GSH were significantly decreased (1.5-fold). Pre-treatment of animals with

P. tomentosa extract apparently attenuated the levels of Lipid peroxidation products and GSH and protected the animals against acetaminophen-induced changes in Lipid peroxidation products and GSH. No significant difference was observed between Group 4 *P. tomentosa* alone treated animals and the control animals (Fig. 1).

Effect of the extract on mitochondrial enzymes

Treatment of animals with acetaminophen markedly decreased the levels of mitochondrial enzymes IDH (1.28-fold), α -KGDH (1.45-fold), SDH (1.92-fold), MDH (1.23-fold), NADH DH (1.44-fold) and cytochrome-C-oxidase (1.59-fold). Whereas pre-treatment with *P. tomentosa* extract protected to a significant level from mitochondrial damage induced by acetaminophen as seen by the near normal levels of mitochondrial enzymes in Group 3 rats. No significant difference was observed between Group 4 *P. tomentosa* alone treated animals and the control animals (Figs. 2–4).

Discussion

Mitochondria are of importance for cell viability due to the nature of its function. It plays a central role in the regulation of intracellular Ca^{2+} especially during toxic insult or stress [24] and mitochondrial oxidative phosphorylation provides approximately 95% of cellular energy needs [25]. Numerous toxic compounds that target mitochondria and mitochondrial injury have been investigated as a potential initiating factor in various organ toxicities caused by chemicals [26]. Since mitochondrial oxidative phosphorylation plays a central role in the maintenance of cellular energy supply [27],

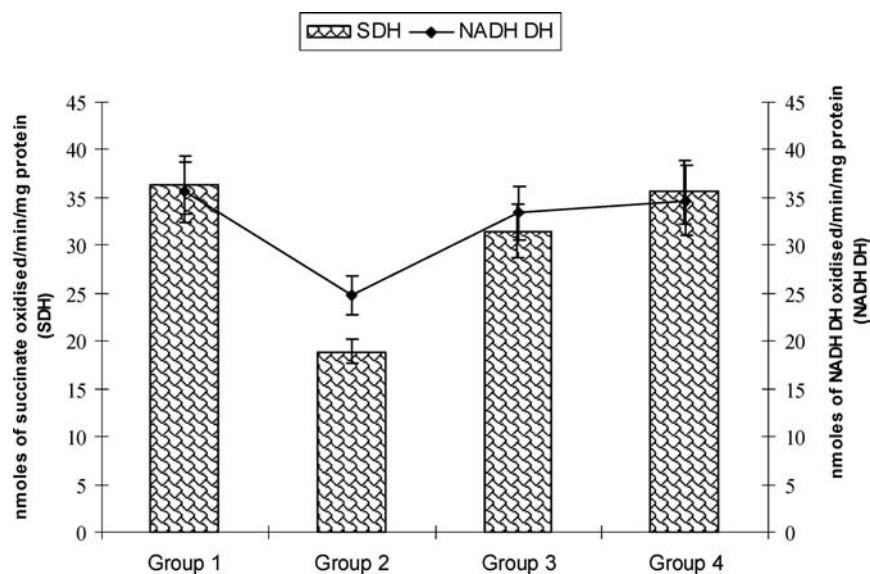


Fig. 2. Statistical comparisons: $p < 0.001$ for Group 2 vs. Group 1 and Group 3 vs. Group 2.

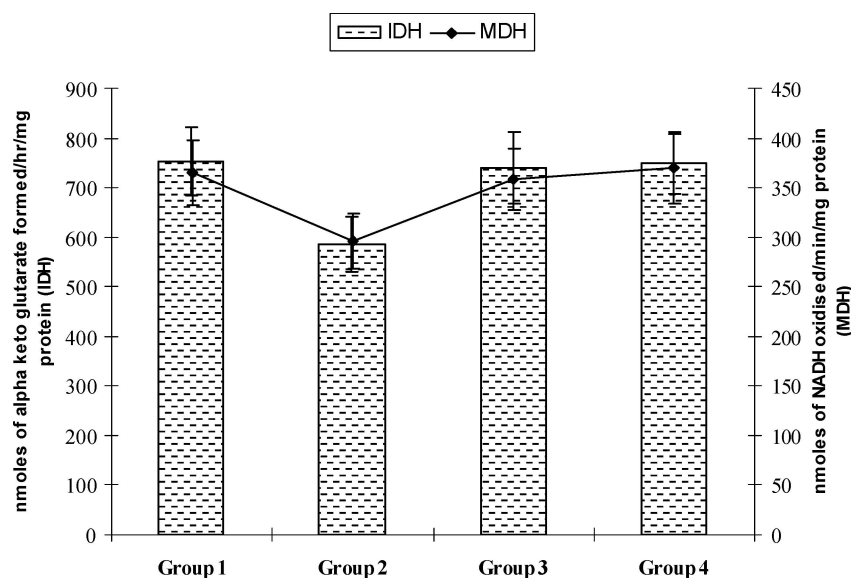


Fig. 3. Statistical comparisons: $p < 0.001$ for Group 2 vs. Group 1; $P < 0.01$ for Group 3 vs. Group 2.

the deleterious effect of xenobiotics on mitochondrial respiration may have serious consequences for the viability of the cell. An example of an early inhibition of mitochondrial respiratory function as a possible initiator of drug induced hepatotoxicity was reported by Donnelly *et al.* [2] who found alterations in mitochondrial oxygen consumption as early as 1 h after administering an hepatotoxic dose of acetaminophen to mice. Since mitochondrial integrity is important for cell viability, the protective effect of the hepatoprotective plant *P. tomentosa* was assessed against acetaminophen-induced hepatotoxicity in rats.

The levels of lipid peroxide products were increased ($p < 0.001$) and cytochrome-C-oxidase were significantly lowered ($p < 0.001$) in acetaminophen-treated Group 2 rats. Lipid peroxidation is recognized as an important deteriorative reaction in biological membranes. The reactive lipid peroxidation by-products such as MDA and increased formation of oxygen-free radicals are known to induce mitochondrial dysfunction [28]. Mitochondrial membrane contains relatively large amounts of polyunsaturated fatty acids in its phospholipid which could probably be the reason for increase in MDA in Group 2 rats. Whereas, pre-treatment with *P. tomentosa*

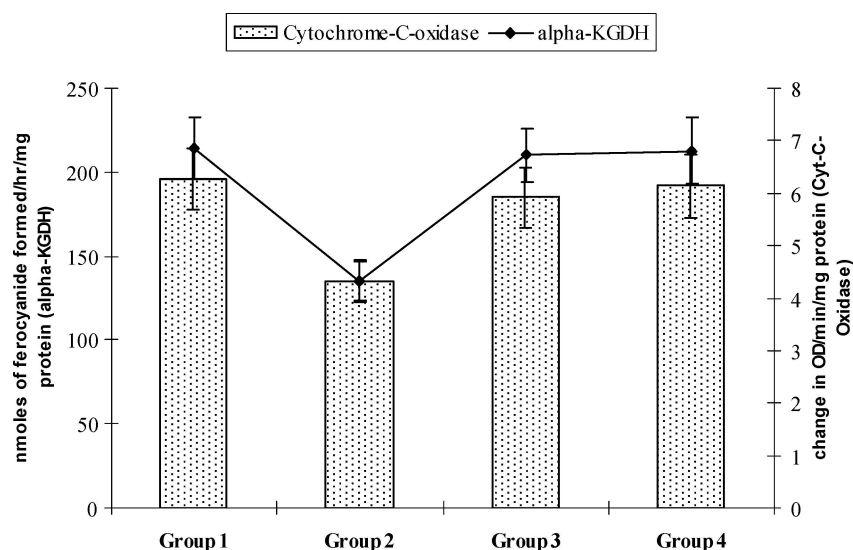


Fig. 4. Statistical comparisons: $p < 0.001$ for Group 2 vs. Group 1 and Group 3 vs. Group 2.

extract was able to sustain the mitochondrial lipid peroxidation products that can be attributed to the defensive action of the extract on oxidative damage [7] by limonene, which removes the agents responsible for peroxidative damage to lipids and aid in the protection against acetaminophen-induced hepatotoxicity.

Cytochrome-C-oxidase, the terminal part of the respiratory chain requires phospholipid for its optimal activity and any change in the lipid composition of mitochondrial membrane is reported to decrease the activity of the enzyme [29]. The increase in Lipid peroxidation products in Group 2 rats could explain the fall in cytochrome-C-oxidase in acetaminophen intoxication. The other reason can be due to oxidative stress induced by acetaminophen, since it is reported that cellular stress produce mitochondrial depletion involving the coding regions of cytochrome-C-oxidase in rat liver [30]. The decrease in the levels of the enzyme by acetaminophen was prevented in rats pre-treated with *P. tomentosa* extract, which might be due to the lowering of lipid peroxidation by the extract.

The levels of mitochondrial GSH were decreased significantly ($p < 0.001$) in acetaminophen intoxicated Group 2 rats. Maintenance of total cellular GSH level is critical for overall preservation of cell homeostasis, however intracellular compartmentalization of GSH has also been shown to be critical in the relationship between GSH depletion and cellular injury [31]. The mitochondrial GSH pool comprises approximately 10% of total hepatic cellular GSH [32] and could conceivably be specifically depleted in response to an hepatotoxic agent, perhaps as a result of local generation of reactive drug metabolites within mitochondria. Depletion of mitochondrial GSH pool would render this

organelle susceptible for oxidative stress which consequently impair cellular respiration and energy production or can alter Ca^{2+} permeability/ Ca^{2+} release from mitochondria with subsequent swelling of the organelle. On the other hand, it can disrupt the cellular function required for detoxification of acetaminophen [33]. Pre-treatment with *P. tomentosa* extract in acetaminophen intoxicated rats resulted in maintenance of GSH levels to near normalcy, probably by stabilizing the Ca^{2+} sequestering ability of mitochondria, thus protecting the respiratory chain proteins against NAPQI, the toxic electrophilic metabolite of acetaminophen. Maintenance of normal levels of GSH could be credited to the presence of limonene in the plant, since dietary limonene (1% for 10 days) is reported to maintain liver GSH concentrations in paracetamol challenged mouse [14].

Acetaminophen administration produced a significant decrease ($P < 0.001$) in the levels of mitochondrial enzymes SDH, IDH, α -KGDH, MDH, NADH dehydrogenase and cytochrome-C-oxidase. Succinate dehydrogenase (SDH) is an important enzyme of TCA cycle and is also associated with the electron transport chain due to its ability to transfer electrons to respiratory chain [34]. SDH is known to contain a number of cysteine rich sulfur clusters and can be inhibited by a number of agents that modify sulfhydryl groups. NAPQI directly interacts with sulfhydryl groups on SDH, causing the loss of its activity [27]. NAPQI also arylates/oxidizes other essential protein sulfhydryl groups in the mitochondrial respiratory chain, thereby limiting the ability of the mitochondria to meet the energy demands of the cell and disrupting cellular energy homeostasis [35]. Pre-treatment with *P. tomentosa* extract prevented the decrease in the levels of SDH and also the levels of IDH, α -KGDH, MDH and NADH dehydrogenase

probably by promoting the elimination of NAPQI by mitochondrial GSH, thereby protecting the critical nucleophilic sites on the enzymes against the toxic electrophilic metabolite NAPQI. The effect of the extract to protect mitochondrial enzymes could be ascribed to the presence of limonene in the plant, since limonene is reported to induce glutathione peroxidase and protect against free radicals by catalysing their inactivation with GSH [36].

Conclusions

The results of the present study show that *P. tomentosa* treatment attenuates mitochondrial damage in acetaminophen-induced hepatotoxicity, by attenuation of peroxidation induced product formation, resulting in the reduction of the development of mitochondrial dysfunction in hepatotoxicity. The management of acetaminophen-induced dysfunction by the extract could be attributed to the presence of D-limonene in the plant which is known to enhance conjugation of toxic metabolites by maintaining liver GSH concentrations.

Acknowledgements

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